

# You Know What I'm Saying: Designing Conversational Strategies of AI Agent for Tip of the Tongue Phenomenon

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## ABSTRACT

The phenomenon of being unable to recall a specific word or term from memory is called “tip-of-the-tongue” (TOT). Despite its significance, potential applications for solving TOT from the user experience (UX) and human-computer interface (HCI) perspectives are yet to be fully explored. With the advent of advanced artificial intelligence (AI) interfaces and technologies, new opportunities have emerged to address TOT. This study aims to investigate the applicability of AI in addressing TOT and the roles of conversation design strategies in mitigating this phenomenon. To this end, we developed a prototype called “Tip-pong,” which is a turn-taking user interface that provides semantic and episodic clues, as well as targeted and optional answers to assist users in resolving TOT. Through a quantitative and qualitative study involving 23 participants, we evaluated the prototype and found that while the type of clue had no significant impact on usability, users preferred optional answers over targeted answers. Additionally, we found that allowing users to experience the process of approaching the answer on their own was more effective than providing the correct answer immediately. Based on these findings, we discuss the design implications for the development of AI agents to solve TOT in everyday life.

## CCS CONCEPTS

• **Human-centered computing** → **Usability testing**.

## KEYWORDS

Conversational agent, tip of the tongue, meta-cognition, priming effect

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## 1 INTRODUCTION

The tip-of-the-tongue (TOT) phenomenon refers to instances in which an individual can remember certain aspects of an object or character but is unable to fully recall its specific details [7]. This phenomenon is common, and many people experience on a daily basis [25], such as recognizing an actor in a TV show or movie but being unable to recall their name. TOT can occur in any type of conversation, whether formal or informal, regardless of the subject matter. Although it may seem insignificant, research has shown that TOT can cause frustration and hinder one's ability to carry on conversations or complete tasks. Although TOT has been studied extensively, there is a need for a deeper understanding of the complexity and variability of TOT in everyday life, as well as for research into effective strategies to overcome it. Extant studies have focused mostly on providing phonological information (initial, accent, etc.) as a way of overcoming TOT [6, 11]. However, these methods may not be practical in real-world situations. Instead of providing peripheral information, it is crucial to understand the situations in which TOT occurs and provide specific strategies or means to solve it.

Recent advancements in AI and ML, coupled with the emergence of new user interfaces such as conversational agents and smart speakers, offer practical approaches to help people cope with TOT. Technologies such as ChatGPT demonstrate the potential of AI to accurately provide users with the information they seek through conversation. These new interfaces or AI agents are easily accessible to users, which helps to avoid situations in which a user may abandon their curiosity and not overcome their TOT. Unlike traditional search engines such as Google, these interactive agents can support users' information retrieval process through conversation, thus reducing the complexity of the process and helping users to narrow down their questions. In this manner, users can obtain with the correct answer immediately, which is consistent with the requirements of immediate and contextualized resolution of TOT.

The aim of this study is to investigate the potential of AI and conversational user interfaces in addressing TOT. To this end, we develop a prototype called “Tip-pong,” which is a conversational user interface that helps users to find correct answers during TOT by providing hints or suggesting possible answers. The prototype provides two different styles of hints (semantic and subjective) and types of answers (targeted one and options) by using the Wizard-of-Oz approach. A user study is conducted in which 23 participants interact with the prototype in a simulated TOT situation and are provided with answers to the questions displayed on the prototype. The results of the evaluation survey indicate that while the

types of clues had no significant effect on usability, the participants unexpectedly prefer optional answers over targeted answers. Additionally, the results of a qualitative study indicate that allowing users to experience the process of approaching the answer on their own is more effective than providing the correct answer immediately because it provides a stronger sense of control and independence. These findings have implications for the design of AI-powered interfaces to address TOT in everyday life.

This study makes an important contribution by introducing interactive AI interfaces to address the TOT phenomenon, which is commonly experienced by people in everyday life but has been largely overlooked outside relevant fields. We present a new approach to TOT that encompasses three key elements: (1) the design of an AI agent and its conversation strategies for addressing TOT, (2) results of a user study that demonstrate how users interact with AI agents to solve TOT problems, as well as the strategies they employ, and (3) design implications for user interfaces for TOT with AI.

## 2 RELATED WORK

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML), along with the emergence of novel user interfaces such as conversational agents and smart speakers [10, 22], offer new and practical ways to help people cope with the TOT. The application of ML algorithms to natural language processing [17], speech recognition, and text-to-speech conversion has led to a level of performance that is almost on par with natural human conversation. The most recent release ChatGPT, a chatbot built using OpenAI's GPT-3 family of large language models, represents the greatest progress in this area with its detailed and articulate responses across multiple knowledge domains. These new interfaces or AI/ML algorithms have several advantages. First, they are easily accessible to users through smart speakers and smartphones and can be used immediately when a user faces a real TOT, thereby helping avoid situations in where a user may decide to abandon their curiosity and not solve the TOT. Second, unlike traditional search engines such as Google, these interactive agents can support users' information retrieval process through conversation, thus reducing the complexity of the process and helping users to narrow down their questions. This can provide users with the correct answers they seek immediately, thus meeting the requirements of immediate and contextualized resolution of TOTs. Additionally, AI agents can be designed to have context-aware capabilities, which allows them to provide more relevant and helpful assistance based on the user's current situation and environment. However, despite these potentials, the applicability of AI/ML and novel interfaces for TOT has not been fully researched yet.

TOT is closely related to metacognition, which is the ability to think about and reflect on one's thinking and learning processes [19]. Individuals experiencing TOT may employ metacognitive strategies such as generating related words and concepts and recalling the context in which they first encountered the desired information. Improved understanding of metacognitive processes in TOT can aid in the identification of strategies to overcome it. However, extant studies have focused on providing snippets of information related to names that people easily forget, such as

providing phonological information to overcome the problem of recalling proper nouns such as human names [7, 15, 21]. A recent study demonstrated that providing semantic or anecdotal information is more effective than providing phonological information alone [5, 6, 11]. Additionally, tracing memories by connecting different words, rather than relying on a single piece of information, can assist in solving TOT. This process can be facilitated through a dialog consisting of questions and answers as the question-asking strategy has been proven to be effective in cognitive science [2, 14]. Furthermore, the anthropomorphism of conversational agents (CAs) can improve the UX, and therefore, human-like expression of inquiries should be considered when designing CAs to respond to uncertain information requests [20].

Meanwhile, various types of memory can be considered in conversation design. Episodic memory refers to the capacity to recollect past events and remember events that occurred in specific spatial and temporal contexts, such as personal experiences [26]. Semantic memory, by contrast, refers to the capacity to recollect facts and general knowledge about the world, regardless of personal experience [26]. While some argue that episodic and semantic memory are dependent on one another [1], others suggest that one should be preceded by the other in terms of importance in memory recollection [9]. Because semantic memories encompass matters that are widely constructed as common knowledge, they can provide accurate and objective information when retrieving answers during TOT. By contrast, episodic memories are useful for re-finding information because contextual details can be capitalized on for recollecting memories. Research has shown that queries that combine context-related and contextual cues outperform semantic queries [13, 16]. However, it is unclear which is superior, and therefore, both types of memory should be reflected in the dialog design for TOT to increase its effectiveness.

Furthermore, the manner in which possible answers to questions are presented should be considered to support users in solving TOT. The classification we note is targeted answers versus list of options. Targeted answers provide a straightforward and specific response to a question without the need to search through additional information. Optional answers, by contrast, provide a list or collection of information related to the question, which allows the user to browse and select the most relevant information. According to an HCI study, compared to only one option in replies, users preferred list-style replies [18]. However, in a specific situation such as when users have limited time or when a recommendation system can predict results with high accuracy, targeted answers can provide users with a better experience. Particularly with AI agents that interact using voice, it may be more challenging to provide multiple types of information simultaneously than is the case with general user interfaces, which makes the provision of targeted information more useful. Therefore, although research on answer type classification is not new, it remains challenging to design an effective approach by context and situation. [12].

In light of this, we attempt to answer the following three research questions in this study: (1) Can continuous conversation with AI user interfaces help solve people's TOT? (2) What strategies can be considered in designing a conversation for solving TOT, specifically in terms of the types of clues and answers? (3) Which elements of user experience are involved in solving TOT with AI user interfaces?

### 3 DESIGN OF TIP-PONG

To answer these research questions, we designed a research probe called "Tip-pong," which is a conversational agent that simulates TOT for users and helps them to solve it. In designing Tip-pong, we aimed to create a conversational user interface that can simulate TOTs and assist users in solving them through conversation. To this end, we considered various factors such as the types of clues and answers and the elements of user experience involved in solving TOT with AI user interfaces. The name Tip-pong was selected to reflect the idea of solving TOT through a dialog, similar to a ping pong game. Tip-pong prototype is a web-based program that runs on a computer, using the Wizard of Oz method [3], where pre-set conversation scripts were input by a researcher. It was operated using a laptop, which was linked to a researcher's laptop for remote control. We designed the conversation corpus through multiple pilot studies to ensure the effectiveness of the prototype in simulating TOT and assisting users in resolving it.

#### 3.1 Celebrity Selection

Our aim was to create an environment in which users could immerse themselves to the extent possible in TOT because it occurs naturally in everyday life. However, it was difficult to replicate TOT in a controlled experimental setting. To replicate TOT as closely as possible, we selected one of the most common types of TOT, that is, the inability to recall proper nouns such as the names of objects or people [15, 21]. Therefore, we set remembering celebrity names as our main experimental task. To design this task, we selected as many celebrities as possible by considering the balance between the major periods in which they were active (before 2000s and after 2000s) and their fields (popular culture, politics, economy, sports, etc.). As a part of our selection process, we conducted a pilot study to include both celebrities that people could easily recall and those that people may have some difficulty recalling. After multiple iterations, we were eventually able to select 120 celebrities for our experiment.

#### 3.2 Simulation of TOT

Tip-pong displayed pictures of each of the 120 celebrities on a screen and asked users questions to determine whether they were experiencing TOT. If the users were unable to identify the celebrity, it was not considered a TOT situation. However, if the users were able to identify the celebrity, Tip-pong would then ask them for the name of the celebrity. If users answered right away "Yes," the situation was not considered a TOT. However, if the users were unable to recall the name and answered "No," the situation was considered a TOT. Once Tip-pong identified a TOT, it continued its conversation with the user by asking further questions, providing hints, and suggesting possible answers to the TOT. In this manner, the prototype was able to simulate TOT and assist users in resolving it.

#### 3.3 Types of Clues and Answers

We designed the conversations of Tip-pong to include two different types of clues and answers (Figure 1).

- Type of Clue (C). According to memory types, the types of conversation between the participants and Tip-pong are divided into semantic cues (SC) and episodic cues (EC). The former refers to the capacity for recollecting facts and general knowledge about the celebrity, such as occupation, gender, nationality, and active period. For example, during the conversation, Tip-pong may ask questions such as "when did you first see this celebrity" or "do you know his/her occupation?" By contrast, EC refers to the capacity for recollecting happenings from the past, for example, remembering events that occurred in particular spatial and temporal contexts. For example, Tip-pong may ask questions such as "what TV show or movie did you see him/her in?"
- Type of answer (A): Tip-pong suggests possible answers that it thinks are close to the correct answer during conversation. The types of answers are divided into targeted answers (TA) and optional answers (OA). A TA represents a single answer (e.g., "It's Emma Watson"), and Tip-pong alone solves the TOT. An OA represents three potentially relevant answers (e.g., "Maybe it's Emma Watson, Daniel Radcliffe, or Rupert Grint") according to the flow of the conversation, and participants can choose from these answers. Figure 1 illustrates the details of the types of clues and answers with sample sentences.

| Semantic Clues + Targeted Answer                                                                                                                                                                                                                                                                                        | Semantic Clues + Optional Answer                                                                                                                                                                                                                                                | Episodic Clues + Targeted Answer                                                                                                                                                                                                                                                     | Episodic Clues + Optional Answer                                                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Hey, Tip-pong, can you help me recall this guy's name?</p> <p>Absolutely. What does he do?</p> <p>He used to be a wrestler, but now he is an actor. An overseas actor.</p> <p>What movie does he appear in?</p> <p>Can't remember the title, but it's about racing. Oh, and he is bald.</p> <p>?</p> <p>TIP-pong</p> | <p>Oh, he's a really famous soccer player, but he's retired now. Maybe.</p> <p>I see. Do you know his nationality?</p> <p>Well, he's a South American. I saw him at the 2002 World Cup. Yellow uniform!</p> <p>Ronaldinho, Ronaldinho, or Rivaldo?</p> <p>?</p> <p>TIP-pong</p> | <p>Oh god, I can't recall anything else. I'm so foolish!</p> <p>When did you see this guy?</p> <p>On Christmas Eve. I always watch this movie by myself at home. The little kid is bothering the thief. It's amusing.</p> <p>Tunt that McClaudy Cullen?</p> <p>?</p> <p>TIP-pong</p> | <p>I can't recall it. He's in my favorite movie.</p> <p>Which scene is on your mind?</p> <p>The British prime minister, flipping a sketchbook... it was cute. Indeed, I also remember the old-fashioned and eccentric singer and manager.</p> <p>Is he among Hugh Grant, Andrew Lincoln, or Bill Nighy from 'Love Actually'?</p> <p>?</p> <p>TIP-pong</p> |

**Figure 1: Four conversation styles of Tip-pong. The conversation styles were created by combining two types of clues (SC/EC) and two types of answers (TA/OA), resulting in four distinct styles: SC+TA, SC+OA, EC+TA, and EC+OA.**

## 4 STUDY DESIGN

To understand how users interact with the system, we designed a user study by following a mixed-methods approach that combines both quantitative and qualitative methods.

### 4.1 Participants and Procedure

To recruit participants, we set the following criteria: People who have experience of using AI-based user interfaces such as VUIs or CAs or those who are interested in them. To ensure the representativeness of our participants, we considered their gender and occupation/major. We first posted a recruitment document on our institution's online community and then used the snowball sampling method. A total of 23 participants were recruited (11 women and 12 men). The ages of the participants ranged from 21 to 32 years ( $M = 29.5$  years;  $SD = 3.3$  years). They reported varying levels of understanding of AI agents (High: 3, Middle: 10, Low: 10

(self-reported)). They were engaged in diverse professions (nurses, startup developers). As compensation for their participation, all participants received a \$10 gift voucher.

Before the experiments began, the participants were provided with an explanation of the AI agent and the TOT phenomenon through a tutorial and informed consent was obtained from them. To enhance their understanding, a short training session was conducted. In the experiment, we used a within-subject design, where participants were required to complete all four conditions of the Tip-pong experiment: SC-TA, SC-OA, EC-TA, and EC-OA. In each condition, Tip-pong would display images of celebrities' faces in sequence and ask participants if they recognized the celebrity. If a TOT was resolved during the conversation, the next celebrity was displayed; otherwise, Tip-pong would keep asking questions, provide semantic or episodic hints, and suggest a targeted answer or optional answers to assist with resolving the TOT. After each session, the participants were administered a five-point Likert scale questionnaire on reliability, intimacy, and satisfaction. Semi-structured interviews were conducted after the completion of all sessions, and the system usability scale (SUS) was evaluated for each experimental condition to evaluate the overall usability of Tip-pong. Throughout the experiment, the participants were asked to perform think-aloud. Each experiment lasted approximately 80 min on average. All sessions were conducted remotely by using Zoom with audio and video recordings.

## 4.2 Analysis

We collected quantitative data from the surveys (SUS and Likert ratings) and qualitative data from the think-aloud sessions and interviews. The SUS scores were analyzed by following the standards used in previous studies. Two-way ANOVA was performed to analyze the Likert ratings of trust, intimacy, and satisfaction under each condition because we used two independent variables (type of response and type of condition). The qualitative data obtained from the think-aloud sessions and post-hoc interviews were transcribed and analyzed using thematic analysis.

## 5 RESULT

This section presents the results of our user study including both quantitative and qualitative data collection methods.

### 5.1 Result 1: Survey

**5.1.1 System Usability Scale (SUS).** The average SUS score of all participants was 81.36. According to Sauro's (2012) standard, it falls in the range of A, indicating that the overall satisfaction level of the users toward the system was within the highest 20% [23]. According to Bangor's (2008) standard, the score was in the "Excellent" range when evaluated using the "adjective range" and "acceptable" when evaluated using the "acceptability range." [4] These results indicated that the participants found Tip-pong "useful" for dealing with TOT, and they were satisfied to a certain degree.

**5.1.2 Evaluation items for Speech response of the Conversational Agent (Two-way ANOVA).** Regarding trust, while the main effect of clue type was insignificant ( $F(1, 88) = 0.83$  and  $p = 1.00$ ), the main effect of answer type was significant ( $F(1, 88) = 4.92$ ,  $p = 0.029$ ,  $n_2 p = 0.053$ ). The reliability of OA ( $M = 4.5$ ,  $SD = 0.55$ ) was significantly

higher than that of TA ( $M = 4.13$ ,  $SD = 0.98$ ), indicating that the participants thought the OA was more reliable than the TA. The interaction effect of clue type and answer type ( $F(1.88) = 0.017$ ,  $p = 0.89$ , and  $n_2 p = 0.00$ ) was insignificant. In terms of intimacy, the main effects of clue type were insignificant ( $F(1, 88) = 0.93$  and  $p = 0.338$ ), and the main effect of answer type was significant ( $F(1, 88) = 4.35$ ,  $p = 0.18$ ,  $n_2 p = 0.01$ ). The intimacy of OA ( $M = 4.3$ ,  $SD = 0.69$ ) was significantly higher than that of TA ( $M = 3.87$ ,  $SD = 1.0$ ). The interaction effect of clue type and answer type was insignificant ( $F(1.88) = 0.23$ ,  $p = 0.63$ ,  $n_2 p = 0.003$ ). In terms of satisfaction, while the main effect of clue type was not significant ( $F(1, 88) = 0.28$ ,  $p = 0.59$ ), the main effects of answer type were significant ( $F(1, 88) = 28.71$ ,  $p = 0.00$ ,  $n_2 p = 0.25$ ). The satisfaction of OA ( $M = 4.5$ ,  $SD = 0.50$ ) was significantly higher than that of TA ( $M = 3.67$ ,  $SD = 0.97$ ). The interaction effect of clue type and answer type was insignificant ( $F(1.88) = 0.07$ ,  $p = 0.79$ ,  $n_2 p = 0.001$ ).

In summary, while the type of clue did not lead to any statistically significant difference in terms of trust, intimacy, and satisfaction, the type of answer did. For all three items, the effects of OA were stronger than those of TA. This result was contrary to our expectation because we believed that TA would be more preferred by participants than OA. These results were confirmed in detail by the results of our qualitative study.

## 5.2 Result 2: Qualitative Analysis

**5.2.1 Impact of Clue Types on the User Experience.** Our first finding pertains to the types of clues that participants desired when attempting to overcome TOT. The survey results revealed that overall, the participants did not have a clear preference for either semantic or episodic clues. Instead, both types of clues had their own advantages and disadvantages for each participant. For example, semantic clues such as nationality or profession provided accurate information and narrowed down possibilities, while episodic clues evoked memories and provided peripheral information that the participants could use. One participant (P2) stated *"When I couldn't recall anything about the celebrity, it was much more useful to have subjective clues than semantic ones. But I prefer a semantic conversation for one clear answer."*

However, some participants clearly preferred one type of clue over the other. Those who preferred semantic clues stated that Tip-pong became more reliable and unbiased when it provided semantic clues based on facts. As P11 stated, *"AI Agents may not be aware of the subjective facts I know because I haven't told them. They will never know my personal standards of humor and beauty."* This shows that users can have skepticism toward or negative thoughts about the agent's episodic characteristics. By contrast, the participants who preferred episodic clues felt that they could remember the person's name faster and had more flexible conversations that helped them solve the TOT. For example, P20 stated *"I frequently misremember factual information. What if he has multiple jobs? What if I can't remember his nationality at all. However, I feel free to express what I remember with episodic clues."*

**5.2.2 Impact of Answer Types on the User Experience.** Our second finding pertains to the type of answer (TA/OA). Overall, the participants expressed relatively high levels of satisfaction with OA when solving TOT. The participants reported that even unrelated

options unexpectedly helped them to find more information about the celebrity and that the elimination of incorrect responses was more beneficial for resolving TOT. P22 stated, *"I had a vague memory of the actor. But when I was presented with Tom Holland and Tobey Maguire as options, I immediately remembered. Even the movie title and plot."* Participants' satisfaction and success levels were significantly higher when they solved TOT by selecting one of their potential responses rather than when Tip-pong provided the TA. P17 said, *"The fact that I remembered and chose the answer gave me a kind of sense of accomplishment."* Notably, most of the participants experienced a certain sense of autonomy in choosing their own answers instead of passively accepting the answers provided by the system. They seemed to feel more satisfied with solving their own questions.

By contrast, the TAs were not highly regarded compared to the OAs, despite users expressing general satisfaction. The main reason for this is that because the TA only provides one answer, it is more likely to be incorrect than the OA. One participant initially had high confidence in the AI agent when the TA was correct, but the confidence level dropped significantly when the TA was incorrect later. Additionally, the TA left the user feeling unfulfilled and demotivated. Users felt that the TA hindered the development of a conversational relationship with the AI agent. A few participants even expressed frustration when the agent quickly provided a correct TA, with comments such as *"I felt like an idiot. I'm sure he knows..."* (P9), and *"It was kinda annoying when Tip-pong suddenly suggested the correct answer."* (P23)

**5.2.3 Feedback after interaction with Tip-pong.** The participants generally expressed contentment with the process of resolving TOT by using Tip-pong and provided suggestions for improvement based on their experiences. They suggested that the system could use previous search and chat histories, as well as everyday speaking habits, to help solve TOT. Personalizing the AI agent in this manner would improve its effectiveness in resolving TOT and increase trust and intimacy between the user and the agent. P14 stated, *"I believe I can solve it more easily if the AI agent is aware of my interests or friends talk."* Additionally, they suggested that TOT should be tailored to each user's specific needs and interests. They believed that their satisfaction with the system would be higher if it was used for their work or interests. P22 said, *"This happens quite often since I'm easily anxious in business meetings. Wouldn't I be able to solve the problem faster if the AI agent was familiar with my major?"* P7 said, *"I think it will be more useful for parents and grandmothers."*

## 6 DISCUSSION

Our findings indicate that a conversation with an AI agent acts as a "prime" to activate participants' memories. The term "priming effect" is used in cognitive science to describe a stimulus that activates a mental concept and influences subsequent behaviors [8]. In TOT, people often have difficulty recalling specific information, even if they are certain they know the information. Therefore, TOT is related to the priming effect in the sense that stimulation of related information can facilitate retrieval of the desired information during TOT. For example, when participants in our experimental task had conversations enumerating "The Devil Wears Prada," and

"Mama Mia," they were more likely to recall "Meryl Streep" successfully than when only one of these movies was mentioned. Thus, AI agents can assist in activating potential memories by facilitating open conversations such as brainstorming and providing a wider range of information and stimuli than users can access on their own.

Additionally, we found that each participant needed a different prime. As the conversation with the AI agent continued, participants' memory types changed in real time, even though the same celebrity image was used. This transition (from object to subject or vice versa) frequently occurs within a conversation. For example, P4 said, *"This is a singer (SC). I saw him at the movies with a friend of mine five years ago (EC). He's the main vocalist (SC), and 'Ay Oh!' comes to mind (EC)."* Therefore, it is important to identify the changes in users' memories in real time and provide a "prime" that is appropriate considering the overall flow of conversation. Also, as indicated by P4's words, participants frequently commented on their personal experiences. They expected that personalized AI agents that consider the users' personal experiences, previous search histories, and interests would be able to provide more accurate information than non-personalized AI agents that provide consistent answers to anyone.

Interestingly, the participants preferred OA over TA according to the quantitative and qualitative findings. For the OA type, finding the correct answer by filtering out incorrect options catalyzed priming and gave the participants a sense of accomplishment. Consequently, the participants clearly desired that Tip-pong serve only as an "assistant" in the TOT solution process. In other words, we confirmed that the participants' self-efficacy grew owing to the metacognitive process of determining answers for themselves from various options [24]. These results enhanced the participants' sense of achievement and satisfaction and created a sense of control over the AI agent.

In conclusion, our results indicate that AI agents can play a significant role in TOT, where individuals struggle to recall specific words or terms from memory. To assist users in overcoming TOT, we suggest the following design considerations for AI agents: (1) Allow users to solve TOT by continuing conversations that provide appropriate questions and answers; (2) provide users with several similar or related options to choose from rather than only one possible answer, which allows them to feel a sense of accomplishment in finding the answer themselves; and (3) identify the user's characteristics and preferences between semantic and episodic memories to present customized hints to each user by combining both types of memories. In the future, we plan to enhance the applicability of our prototype to real-life scenarios by incorporating state-of-the-art AI technology such as ChatGPT. This will enable users to use the prototype in their daily lives when they encounter TOT and help them to retrieve information more easily. Additionally, we aim to expand the diversity of situations and user groups, such as elderly individuals, to broaden the reach of our results. By analyzing the outcomes, we hope to identify specific use cases in which our system can be the most helpful and develop conversation strategies tailored to each case.

## REFERENCES

- [1] Jaime Arguello, Adam Ferguson, Emery Fine, Bhaskar Mitra, Hamed Zamani, and Fernando Diaz. 2021. Tip of the tongue known-item retrieval: A case study in movie identification. In *Proceedings of the 2021 Conference on Human Information Interaction and Retrieval*. 5–14.
- [2] Shiho Arima, Shingo Kuroiwa, Yasuo Horiuchi, and Daisuke Furukawa. 2015. Question-asking strategy for people with aphasia to remember food names. *Journal on Technology & Persons with Disabilities* 3 (2015), 10–19.
- [3] Jonathan Arnowitz, Michael Arent, and Nevin Berger. 2007. *Wizard-of-oz Prototyping*. 402–419. <https://doi.org/10.1016/B978-012088568-8/50022-9>
- [4] Aaron Bangor, Philip T Kortum, and James T Miller. 2008. An empirical evaluation of the system usability scale. *Intl. Journal of Human–Computer Interaction* 24, 6 (2008), 574–594.
- [5] Serge Brédart and Marie Geurten. 2020. Strategies to resolve recall failures for proper names: New data. *Memory & Cognition* 48, 8 (2020), 1417–1428.
- [6] Tim Brennen, Thom Baguley, Jim Bright, and Vicki Bruce. 1990. Resolving semantically induced tip-of-the-tongue states for proper nouns. *Memory & Cognition* 18, 4 (1990), 339–347.
- [7] Roger Brown and David McNeill. 1966. The “tip of the tongue” phenomenon. *Journal of verbal learning and verbal behavior* 5, 4 (1966), 325–337.
- [8] Kahneman Daniel. 2017. Thinking, fast and slow.
- [9] David Elswiler, Ian Ruthven, and Christopher Jones. 2007. Towards memory supporting personal information management tools. *Journal of the American Society for Information Science and Technology* 58, 7 (2007), 924–946.
- [10] Teresa Fernandes and Elisabete Oliveira. 2021. Understanding consumers’ acceptance of automated technologies in service encounters: Drivers of digital voice assistants adoption. *Journal of Business Research* 122 (2021), 180–191.
- [11] J Richard Hanley and Elaine S Cowell. 1988. The effects of different types of retrieval cues on the recall of names of famous faces. *Memory & Cognition* 16, 6 (1988), 545–555.
- [12] Tianyong Hao, Wenxiu Xie, Qingyao Wu, Heng Weng, and Yingying Qu. 2017. Leveraging question target word features through semantic relation expansion for answer type classification. *Knowledge-Based Systems* 133 (2017), 43–52.
- [13] Liadh Kelly, Yi Chen, Marguerite Fuller, and Gareth JF Jones. 2008. A study of remembered context for information access from personal digital archives. In *Proceedings of the second international symposium on Information interaction in context*. 44–50.
- [14] Kazuhiro Kuwabara, Takayuki Iwamae, Yudai Wada, Hung-Hsuan Huang, and Keisuke Takenaka. 2016. Toward a conversation partner agent for people with aphasia: Assisting word retrieval. In *International Conference on Intelligent Decision Technologies*. Springer, 203–213.
- [15] Ye Ni Lee, So Young Choi, Ye Ni Lee, and So Young Choi. 2016. Differences in tip-of-the-tongue phenomenon and resolving patterns according to vocabulary characteristics between young and elderly adults. *Communication Sciences & Disorders* 21, 1 (2016), 139–150.
- [16] Q Vera Liao, Biplav Srivastava, and Pavan Kapanipathi. 2017. Tailoring Conversational UX through the Lens of Dialogue Complexity. In *CHI Workshop on Conversational UX Design*.
- [17] Junfeng Liu, Christopher Symons, and Ranga Raju Vatsavai. 2022. Persona-Based Conversational AI: State of the Art and Challenges. *arXiv preprint arXiv:2212.03699* (2022).
- [18] Qianli Ma, Yaping Zhang, Wenti Xu, and Ronggang Zhou. 2022. Ask a Further Question or Give a List? How Should Conversational Agents Reply to Users’ Uncertain Queries. *International Journal of Human–Computer Interaction* (2022), 1–15.
- [19] Janet Metcalfe, Bennett L Schwartz, and Paul A Bloom. 2017. The tip-of-the-tongue state and curiosity. *Cognitive research: principles and implications* 2, 1 (2017), 1–8.
- [20] Clifford Nass and Youngme Moon. 2000. Machines and mindlessness: Social responses to computers. *Journal of social issues* 56, 1 (2000), 81–103.
- [21] Costanza Papagno and Erminio Capitani. 1998. Proper name anomia: A case with sparing of the first-letter knowledge. *Neuropsychologia* 36, 7 (1998), 669–679.
- [22] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, Peter J Liu, et al. 2020. Exploring the limits of transfer learning with a unified text-to-text transformer. *J. Mach. Learn. Res.* 21, 140 (2020), 1–67.
- [23] Jeff Sauro and James R Lewis. 2016. *Quantifying the user experience: Practical statistics for user research*. Morgan Kaufmann.
- [24] Bennett L Schwartz and Alan S Brown. 2014. *Tip-of-the-tongue states and related phenomena*. Cambridge University Press.
- [25] Bennett L Schwartz and Janet Metcalfe. 2011. Tip-of-the-tongue (TOT) states: Retrieval, behavior, and experience. *Memory & cognition* 39, 5 (2011), 737–749.
- [26] Larry R Squire and Stuart M Zola. 1998. Episodic memory, semantic memory, and amnesia. *Hippocampus* 8, 3 (1998), 205–211.